

SEARCHING

Q4)

Question:

DS	Session	Searching	Question Information	Level 1 Challenge 4
Question Description: Ragu has given a range [L, R] to Smith. Smith wants to require to find the number of integers 'X' in the range such that $\text{GCD}(X, F(X)) > 1$ where $F(x)$ is equal to the sum of digits of 'X' in its hexadecimal (or base 16) representation. Example : $F(27) = 1+B=1+11=12$ (27 in hexadecimal is written as 1B) Constraints: $1 \leq T \leq 50$ $1 \leq L$ $R \leq 10^5$ Input Format: The first line contains a positive integer 'T' denoting the number of questions that you are asked. Each of the next 'T' lines contain two integers 'L' and 'R' denoting the range of questions.				

PROGRAM:

```
1  #include <stdio.h>
2
3  int gcd(int a,int b) {
4      while(b) {
5          int t=a%b;
6          a=b;
7          b=t;
8      }
9      return a;
10 }
11
12 int hexsum(int n) {
13     int s=0;
14     while(n) {
15         s+=n%16;
16         n/=16;
17     }
18     return s;
19 }
20
21 int main() {
22     int p[100001]={0};
23     int t,l,r,i,x;
24
25     for(i=1;i<=100000;i++) {
26         x=hexsum(i);
27         p[i]=p[i-1]+(gcd(i,x)>1);
28     }
29
30     scanf("%d",&t);
31
32     while(t--) {
33         scanf("%d%d",&l,&r);
34         printf("%d\n",p[r]-p[l-1]);
35     }
36
37     return 0;
38 }
```

OUTPUT:

TEST CASE 1:

```
3
1 3
2
5 8
4
7 12
6
```

TEST CASE 2:

```
10
15 64
29
14 100
52
9 79
45
19 63
26
3 83
53
5 61
37
18 93
45
20 98
47
13 71
36
12 53
26
```